UAT

Gummadidala Geetha Charani

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2 a) Theorem (Universal Approximation Theorem).

Let $K \subset \mathbb{R}^n$ be a compact set and let $f \in C(K)$ be a continuous function on K. Let $\sigma : \mathbb{R} \to \mathbb{R}$ be a sigmoid activation function.

Then for every $\varepsilon > 0$, there exists a feedforward neural network with a single hidden layer and finitely many hidden neurons N, with parameters $\{a_i, w_i, b_i\}_{i=1}^N$, such that the function

$$F(x) = \sum_{i=1}^{N} a_i \, \sigma(w_i^T x + b_i)$$

satisfies

$$\sup_{x \in K} |f(x) - F(x)| < \varepsilon.$$

The approximation is in the uniform (supremum) norm, defined as

$$\|f - F\|_\infty = \sup_{x \in K} |f(x) - F(x)|.$$

- **Class of Target Functions:** $f \in C(K)$, i.e., the set of all continuous functions defined on a compact set $K \subset \mathbb{R}^n$.
- **Assumptions on the Domain:** The domain K is compact (closed and bounded)

  The input space is finite-dimensional $\mathbf{R^n}$.

b) Main Idea Behind the Universal Approximation Theorem

The main idea of the proof is that a single hidden layer neural network can construct flexible nonlinear basis-like functions whose linear combinations can approximate any continuous function on a compact domain.

Each hidden neuron computes a function of the form

$$\sigma(w_i^T x + b_i),$$

where σ is a continuous, non-polynomial activation function- the sigmoid function. By varying the parameters w_i and b_i, these functions can be shifted, scaled, and oriented in the input space.

The overall network output is a linear combination of such functions:

$$F(x) = \sum_{i=1}^{N} a_i \, \sigma(w_i^T x + b_i).$$

The key theoretical argument is that the collection of functions of this form is *dense* in $C(K)$, the space of continuous functions on a compact set $K \subset \mathbb{R}^n$, with respect to the uniform (supremum) norm. This means that for every continuous function f and every $\varepsilon > 0$, there exists a finite number of neurons N and parameters $\{a_i, w_i, b_i\}_{i=1}^N$ such that

$$\sup_{x \in K} |f(x) - F(x)| < \varepsilon.$$

The proof connects neural networks to classical approximation theory, particularly the Stone–Weierstrass theorem, which states that certain families of functions are dense in $C(K)$. Because the activation function is continuous and non-polynomial, the set of neural network functions is rich enough to satisfy this density property.

Intuitively, each hidden neuron acts like a smooth step or bump function, and by combining sufficiently many such functions, the network can approximate arbitrary continuous functions uniformly over compact domains.

c) Limitations of the Universal Approximation Theorem

Although the Universal Approximation Theorem (UAT) guarantees that a single hidden layer neural network can approximate any continuous function on a compact domain, it has several important limitations from a practical machine learning perspective.

1. The theorem is an existence result, not a training guarantee. The UAT states that there exists a neural network with sufficiently many hidden neurons that can approximate a given function arbitrarily well. However, it does not guarantee that standard training algorithms (such as gradient descent or backpropagation) will find the required parameters.

In practice, neural network training involves solving a highly non-convex optimization problem. The loss surface may contain local minima, saddle points, or flat regions. As a result, even though a suitable network representation exists, the optimization procedure may fail to converge to it.

This limitation is relevant because successful approximation in theory does not imply successful learning in practice.

2. The theorem does not specify the required network size. The UAT guarantees approximation with a finite number of hidden neurons, but it does not provide any bound on how large the network must be. In some cases, achieving a good approximation with a single hidden layer may require an extremely large number of neurons.

Such large networks can lead to:

- High computational cost,
- Overfitting,
- Increased memory usage,
- Difficult training dynamics.

In practice, this is why deep networks (with multiple hidden layers) are often preferred, as they can represent complex functions more efficiently than very wide shallow networks.

d) Universal Approximation with ReLU and Increased Depth

When using ReLU activations and allowing network depth to increase while keeping the width relatively small, the nature of the Universal Approximation property changes in an important way.

First, ReLU ($\sigma(x) = \max(0, x)$) is a continuous, non-polynomial activation function. Therefore, even a single hidden layer ReLU network retains the Universal Approximation property: it can approximate any continuous function on a compact set arbitrarily well, provided it has sufficient width.

However, when depth is allowed to increase while keeping width relatively small, neural networks can represent complex functions much more efficiently. Deep ReLU networks can approximate certain classes of functions using exponentially fewer neurons than shallow networks. In other words, depth can trade off with width in terms of expressive power.

From a practical perspective, this means:

- Shallow networks require very large width to approximate complex functions.
- Deep networks can achieve similar approximation accuracy with significantly fewer neurons per layer.
- Increasing depth improves expressive efficiency, even though the universal approximation property already holds for shallow networks.

Thus, while universal approximation does not fundamentally change with ReLU activations, allowing greater depth makes function approximation far more efficient and practical.